Introduction

In the previous chapter, discussions were on various types of data, census and sampling and methods of collecting data with suitable examples. It may be recalled that by data we mean the collected set of a number of related observations. The data as such are complex and voluminous in nature in many instances. It is essential that the data so collected must be arranged for proper understanding and for further statistical analysis.

3.1 Classification of Data



Few 1000 years ago the ancient Tamil poet Tholkappiyar classified things on earth into two categories as living and non-living things. He also classified the living beings based on their six senses.

The data that are unorganized or have not been arranged in any way are called raw data. The ungrouped data are often voluminous, complex to handle and hardly useful to draw any vital decisions. Hence, it is essential to rearrange the elements of the raw data set in a specific pattern. Further, it is important that such data must be presented in a condensed form and must be classified according to homogeneity for the purpose of analysis and interpretation. An arrangement of raw data in an order of magnitude or in a sequence is called **array**. Specifically, an arrangement of observations in an ascending or a descending order of magnitude is said to be an **ordered array**.

Classification is the process of arranging the primary data in a definite pattern and presenting in a systematic form. *Horace Secrist* defined classification as the process of arranging the data into sequences and groups according to their common characteristics or separating them into different but related parts. It is treated as the process of classifying the elements of observations or things into different groups or classes or sequences according to the resemblances and similarities of their character. It is also defined as the process of dividing the data into different groups or classes which are as homogeneous as possible within the groups or classes, but heterogeneous between themselves.

Objectives of Classification

Classification of data has manifold objectives. The salient features among them are the following:

- It explains the features of the data.
- It facilitates comparison with similar data.
- It strikes a note of homogeneity in the heterogeneous elements of the collected information.
- It explains the similarities which may exist in the diversity of data points.
- It is required to condense the mass data in such a manner that the similarities and dissimilarities are understood.

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